PHISHING EMAIL CHECKER

Team Task Allocation and Project Closeout Plan

Project stage	Testing, Validation and Final Submission
Team size	4 members

PROJECT OBJECTIVE

The application has already been implemented. The remaining work focuses on validating its functionality, evaluating its effectiveness, reviewing security and privacy, preparing documentation, and producing a complete final submission.

Each team member is responsible for completing their assigned tasks and submitting the specified deliverables.

The allocation separates ownership across application integration, detection testing, security assurance, and documentation/presentation so that all four members have clear areas of accountability.

PERSON 1 — TECHNICAL LEAD AND APPLICATION INTEGRATION

Objective

Ensure that the completed application is functional, properly integrated, secure and ready for final assessment.

Tasks

1. Run the frontend and backend and verify that both operate correctly.
2. Test communication between the frontend and backend.
3. Verify email submission, analysis results, risk scores, detected indicators and recommendations.
4. Test analysis history, individual record retrieval and history deletion.
5. Review API input validation and error handling.
6. Check CORS configuration and ensure API credentials and environment variables are not exposed.
7. Review SQLite storage and resolve the privacy concern arising from storing submitted email content.
8. Verify that suspicious URLs are never automatically visited or fetched.
9. Run the backend test suite and frontend production build.
10. Fix critical technical issues identified during the final review.
11. Check the GitHub repository for exposed secrets, unnecessary files and missing dependencies.
12. Confirm that the project can be installed and run using the documented instructions.

Required deliverables

• Verified working application.

• Technical and privacy review.

• Backend test results and frontend build verification.

• Clean and organised GitHub repository.

• Final technical readiness report.

• Confirmation that the application is ready for demonstration.

PERSON 2 — TEST DATASET DEVELOPMENT AND DETECTION TESTING

Objective

Evaluate the phishing detection engine using a structured collection of fictional email samples.

Tasks

1. Create a labelled dataset containing benign, suspicious and ambiguous emails.
2. Provide an expected classification and explanation for every sample.
3. Test detection of urgent and threatening language; requests for passwords, OTPs and payment information; suspicious URL shorteners; IP-address-based URLs; lookalike domains; sender display-name impersonation; and unexpected attachment references.
4. Test edge cases: empty submissions, extremely long email content, malformed URLs, Unicode and unusual characters, harmless urgent messages, suspicious-looking words in otherwise legitimate messages, and missing sender information.
5. Compare expected classifications with actual application results.
6. Record correctly detected indicators and incorrect classifications.
7. Identify false positives and false negatives.
8. Document detection weaknesses and recommend improvements.

Required deliverables

- Labelled fictional email dataset.
- Test-case documentation.
- Completed test-results table.
- False-positive and false-negative analysis.
- Detection effectiveness summary.
- Recommendations for improving detection.

Important: Do not use real people's private emails or visit suspicious links. Do not present results from fictional samples as proof of real-world detection accuracy.

PERSON 3 — SECURITY REVIEW AND QUALITY ASSURANCE

Objective

Conduct an independent security and quality review of the application, identify potential weaknesses and verify that the application behaves safely under different conditions.

Tasks

1. Review the application from a security perspective.
2. Examine how the application handles submitted email content and URLs.
3. Verify that suspicious URLs are analysed as text without automatically accessing external websites.
4. Check that user input is appropriately validated and displayed safely.
5. Review error handling for malformed or unexpected requests.
6. Examine whether email content, sender information or other sensitive data is unnecessarily exposed through logs or stored records.
7. Verify that environment variables, API keys and other secrets are not exposed in the repository or frontend.
8. Review access to analysis history and deletion functionality.
9. Test the API using its documentation interface: valid analysis requests; empty or incomplete requests; invalid input formats; nonexistent analysis record IDs; and history retrieval and deletion.
10. Record any security or quality issues discovered.
11. Assign a priority to each issue based on its potential impact.
12. Recommend practical remediation measures and verify fixes where possible.

Required deliverables

- Security and quality assurance report.
- API validation results.
- List of identified issues and their severity reasoning.
- Evidence supporting reported issues.
- Recommended remediation measures.
- Verification of resolved issues.

Important: This is an authorised review of the team's own application. Do not conduct testing against external websites, email systems or infrastructure without permission.

PERSON 4 — DOCUMENTATION, EVIDENCE AND PRESENTATION

Objective

Prepare the written documentation, visual evidence and presentation materials required for the final project submission.

Tasks

1. **Prepare the README.** Include project title and description, problem statement, target users, main features, technologies, project structure, installation instructions, backend and frontend startup instructions, API documentation access, test instructions, a fictional example, privacy considerations and limitations.
2. **Prepare the technical report.** Include a cover page, executive summary, problem statement, objectives, target audience, scope and limitations, system architecture and methodology, technologies, implementation overview, testing methodology, results and findings, recommendations, conclusion and appendices.
3. **Describe significant findings.** For each finding, include its title, affected component, supporting evidence, safe reproduction steps, potential impact, severity reasoning where applicable, and recommended remediation.
4. **Capture project evidence.** Take clear screenshots of the main dashboard, email submission interface, suspicious and benign email analyses, risk scores, detected indicators, recommendations, history where appropriate, API documentation and test execution results.
5. **Prepare presentation slides.** Cover project title and team members, problem statement, why phishing detection matters, objectives and target users, scope and limitations, solution overview, architecture, technologies, features, testing methodology and results, security and privacy considerations, recommendations, conclusion and individual contributions.
6. **Prepare the demonstration script.** Show how a user submits an email, how the application analyses it, how risk indicators are displayed, how safe next steps are recommended, and how history can be reviewed and cleared if enabled.
7. **Prepare the team contribution summary.** Document each member's assigned responsibilities and completed contributions.

Required deliverables

- Complete README.
- Technical project report.
- Application screenshots and supporting evidence.
- Presentation slides.
- Demonstration script.
- Team contribution summary.

5. TEAM-WIDE RESPONSIBILITIES

All four members must:

- Complete assigned tasks within the agreed timeframe.
- Communicate progress and report challenges early.
- Submit their deliverables for review.
- Keep project contributions organised in the shared repository where appropriate.
- Participate in the final review and presentation preparation.
- Ensure all evidence is accurate and reproducible.
- Review the final submission package before submission.

6. FINAL SUBMISSION CHECKLIST

SUBMISSION ITEM	PRIMARY RESPONSIBILITY
Working application and clean repository	Person 1
Technical and privacy review	Person 1
Labelled test dataset	Person 2
Detection testing and evaluation report	Person 2
Security and quality assurance report	Person 3
API validation evidence	Person 3
README and technical report	Person 4
Screenshots and supporting evidence	Person 4
Presentation slides and demo script	Person 4
Team contribution summary	Person 4, confirmed by all members
Final quality review and submission approval	All members, coordinated by Person 1

7. INDIVIDUAL PROGRESS REPORTING

Each member must submit a brief progress update containing:

- **Assigned responsibilities:** Tasks allocated to the member.
- **Completed tasks:** Tasks successfully finished.
- **Deliverables produced:** Documents, reports, screenshots, test results or code contributions.
- **Outstanding tasks:** Work that remains incomplete.
- **Challenges:** Issues requiring assistance.
- **Completion status:** Completed, in progress or pending.

8. FINAL EXPECTATION

The team is expected to deliver a working, tested, secure, documented and demonstrable Phishing Email Checker that satisfies the original project proposal and programme closeout requirements.

Every member is accountable for their assigned deliverables. The technical lead coordinates integration and final quality approval, but all four members are expected to contribute meaningfully to the successful completion of the project.